

Planar Coils for Detection of Magnetic Nanoparticles in a Testbed for Molecular Communication

Max Bartunik*

max.bartunik@fau.de

Institute for Electronics Engineering,
Friedrich-Alexander-Universität Erlangen-Nürnberg (FAU)
Erlangen, Germany

Timo Maiwald

Institute for Electronics Engineering,
Friedrich-Alexander-Universität Erlangen-Nürnberg (FAU)
Erlangen, Germany

Samira Faghih-Naini*

samira.sf.faghih-naini@fau.de

Institute for Electronics Engineering,
Friedrich-Alexander-Universität Erlangen-Nürnberg (FAU)
Erlangen, Germany

Jens Kirchner

Institute for Electronics Engineering,
Friedrich-Alexander-Universität Erlangen-Nürnberg (FAU)
Erlangen, Germany

ABSTRACT

Testbeds enable experimental investigation of molecular communication concepts and transmission schemes. One approach to bio-compatible molecular communication, implemented in a testbed, is the use of magnetic nanoparticles as information carriers. These nanoparticles can be detected based on their high susceptibility using an inductive sensor. So far, a wire-wound sensing coil, that requires manual manufacturing and is difficult to install for a real-life transmission channel, was used. Here, we propose the use of planar coils. To this end, a selection of commercially available coils are evaluated in comparison to a conventional wire-wound coil as reference. A planar coil is then successfully used to detect magnetic nanoparticles in the context of the molecular communication testbed. The contribution is concluded with an outlook on future improvements made possible through the application of planar sensing coils.

CCS CONCEPTS

• **Hardware** → **Sensor devices and platforms.**

KEYWORDS

Molecular communication, Testbed, Magnetic nanoparticles

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*Both authors contributed equally to this paper

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1 INTRODUCTION

In molecular communication, molecules or similarly small particles are used to transmit information in environments where conventional electromagnetic communication is either impossible or infeasible [10]. Some typical use cases are therapeutics or diagnostics in medicine [11].

Testbed setups are essential to investigate the possibilities and limitations of molecular communication in real-life scenarios. Although a variety of testbeds have been published, very few setups have a biocompatible design, which is essential for the context of medical applications.

One of the first tabletop demonstrators for molecular communication was presented in [8] and uses alcohol to transmit information in air. Various derivations of this work, such as the extension to a MIMO setup [15], have also been published. Other air-based setups use fluorescent particles [7] or odors [12, 17] as information carriers. However, we will concentrate on fluid-based testbeds in the following as air-based setups are not applicable to the biomedical use-case.

Fluid-based testbeds typically use an active background flow. Setups where information is modulated as pH value [9, 13], conductivity with saline solution [22], glucose concentration [14] or as a pharmacokinetic response, modeled using experimental medical data [1], have been presented.

We will be using a testbed based on biocompatible magnetic nanoparticles (MNPs), specifically superparamagnetic iron-oxide nanoparticles [4, 5, 21, 23]. In contrast to setups with other information carriers, MNPs do not occur naturally in biomedical environments and can therefore be applied with a minimal amount of background noise. Inductive sensors with wire-wound coils were used as a receiver for MNPs in previous work [4, 21]. However, such coils have to be adjusted to the specific setup and are difficult to manufacture. In addition, in a real-life application it may not be possible to separate the transmission channel (e.g. a blood vessel) to pass it through a sensor coil. We therefore propose the use of planar sensor coils.

Planar coils have been used for multiple use-cases, such as wireless power transfer. In this context possible shapes and inductive behaviour have been investigated [16, 19]. Furthermore, planar coils have been successfully used to manipulate ferrofluids [18] and propel microbeads for lab analysis [20].

In this work we will present a new sensor design for inductive sensing of MNPs in a molecular communication context. A variety of planar coils are evaluated for the given use-case in the context of an experimental testbed. The ideal sensor coil should have a high sensitivity to the used MNPs and a narrow impulse response. The proposed coils are commercially available and easy to implement for a custom setup on a printed circuit board. Some of the disadvantages of the previous approach using wire-wound coils, such as the need to pass the transmission channel through the sensor coil and an imprecise manual manufacturing procedure, are eliminated with planar coils, placed on the outside of the transmission channel.

2 TESTBED CONCEPT

The testbed can be separated into the three components transmitter, channel and receiver. The transmitter, a peristaltic pump (Ismatec Reglo ICC Digital Pump), injects a suspension of MNPs (manufactured by the Section for Experimental Oncology and Nanomedicine of the University Hospital Erlangen (SEON)) into the transmission channel. The used suspension has a stock iron concentration of 7.5 mg mL^{-1} . Injections are modulated to achieve on-off-keying.

The channel consists of tubing with an inner diameter of 1.52 mm and contains a constant background flow of water with a flow rate of 10 mL min^{-1} also driven by a peristaltic pump. The flow inside the tube is laminar, causing an axial distribution of an injected nanoparticle burst. The concentration of MNPs observed over time at the receiver typically has a steep rising edge and a slow return to zero.

Sensing at the receiver (typically at a distance of 5 cm from the point of injection) is based on the MNPs magnetic properties by detecting changes in susceptibility with a coil. Custom made devices or a commercial susceptometer with wire wound coils were used in previous work [4, 6, 21]. Appropriate wire wound coils cannot be obtained of the shelf, are difficult to manufacture consistently, and must be wound around the transmission channel. We therefore employed printed planar coils, described in Section 3.2, for the receiver.

Data transmission, in the context of molecular communication, is achieved by modulating the nanoparticle bursts using on-off-keying or concentration-shift-keying. As shown in [2] data rates of up to 5.5 bit s^{-1} can be achieved in this manner.

Fig. 1 provides an overview of the whole setup. Two independently controlled channels of the peristaltic pump are used to provide both the constant water flow and the modulated MNP injections. The injection is achieved with a y-connector placed against the direction of flow to reduce washing out of nanoparticles from the injection tube.

3 SENSOR

The inductive sensor LDC1614 from Texas Instruments is used as a receiver. The four channel sensor drives a resonance circuit with an ac signal to determine its resonance frequency. In our setup a parallel circuit with the sensor coil and a capacitor is used. The resonance sensing range is 1 kHz to 10 MHz. As MNPs pass through or by the coil, their susceptibility χ causes a shift in the inductance L of the coil, which in turn changes the resonance frequency f_{res}

of the parallel circuit according to

$$\Delta f_{\text{res}} = f_{\text{res}}(L_2) - f_{\text{res}}(L_1) = \frac{\sqrt{L_1} - \sqrt{L_2}}{2\pi\sqrt{L_1 L_2 C}}, \quad (1)$$

with the capacitance of the resonance circuit C .

3.1 Wire-Wound Coils

In previous work, the LDC1614 or the similar model LDC1612 with only two sensing channels, was used in combination with wire-wound coils [4, 5]. These coils were manufactured specifically to match the geometry of the transmission channel. They have a length of 20 mm, an inner diameter of 4 mm to accommodate the transmission channel and consist of 20 turns. Each coil was manually wound using bondable enamelled wire and then set with hot air to ensure mechanical stability. Such a coil will be used as reference in the following.

3.2 Planar Sensor Coils

Like wire-wound coils, planar coils also produce a magnetic field when excited. However, while the magnetic field of a coil that is long relative to its core diameter is strongly focused within the core of the coil, a planar structure (comparable to a short coil with many layers) creates a relatively strong magnetic field on the surface of the coil. Fig. 2 visualises the magnetic field distribution qualitatively for such a planar coil with an outer diameter of 13 mm using the simulation tool "Finite Element Method Magnetics"¹.

To detect MNPs in the transmission channel, the planar coil is attached to the transmission channel with adhesive tape. Shifts in resonance frequency, caused by the MNPs, are then detected using the LDC1614 inductive sensor and evaluated using deep learning to demodulate the transmission (see [3] for details on the demodulation process).

The variable parameters of the investigated planar coils are the amount of layers, the trace width and spacing, and the inner and outer diameters of the coil. The amount of turns results from the other parameters. The coils (part of LDCCOILEVM from Texas Instruments), have up to four layers printed on an appropriate circuit board (PCB). Coils with two layers have one layer each on the outer surfaces of the PCB. An overview of the evaluated coils is provided in Table 1. The planar coils are denoted as 'A' through 'F', the wire-wound coil used as reference as 'Ref'.

4 EVALUATION

The various coils are evaluated in the context of the testbed. An ideal sensor coil shows a large inductance shift when exposed to MNPs with a low level of background noise. Small coil sizes are preferable, as they allow for a compact sensor design. A planar coil design should match or exceed the performance of the conventional reference coil.

4.1 Coil Response

First, the coils are evaluated independent of the inductive sensor. To this end, the inductance of the coils at a frequency of 500 kHz, below the lowest self-resonant frequency, is measured directly using the impedance analyser 4194A from HP.

¹<https://www.femm.info>



Figure 1: Overview of the testbed setup. Information is transmitted through modulated injections of MNPs into the transmission channel with a constant background flow of water. MNPs are detected at the receiver using a planar coil.

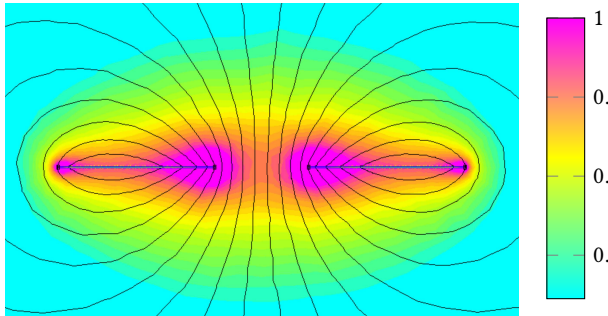


Figure 2: Magnetic field distribution for the cross section of a planar coil with an outer diameter of 13 mm. The colouring corresponds to the magnetic field strength, normalised to the highest intensity, which is observed at the inner edge of the coil.

The transmission channel, attached to the coil under investigation, is first filled with water and then, in a second step, replaced with a solution of MNPs. The inductance is recorded for both states.

The first four columns of Table 2 show the inductance value L measured for a channel filled with water and the absolute (ΔL) and relative ($\Delta L/L$) inductance shift observed when introducing the MNP solution. As expected, the inductance increases with the susceptibility of the sensor target. The relative inductance shift for all coils is below 0.01 %, with significant differences for the various

Table 1: Overview of the relevant properties of the evaluated planar coils 'A' through 'F' and the wire-wound coil 'Ref' used as baseline reference.

Coil	Diameter	Turns	Trace/Wire		
			Width	Spacing	Layers
Ref	6 mm	20	1.00 mm	-	20
A	46 mm	100	0.15 mm	0.15 mm	2
B	46 mm	160	0.20 mm	0.15 mm	4
C	29 mm	70	0.15 mm	0.15 mm	2
D	29 mm	120	0.20 mm	0.15 mm	4
E	13 mm	96	0.10 mm	0.10 mm	4
F	8 mm	44	0.10 mm	0.10 mm	4

planar coil geometries. The maximal relative shift is achieved with the planar coil 'B'.

4.2 Sensitivity

In a next step, the coils were evaluated regarding their sensitivity in the complete sensor setup. As a resonance circuit is necessary, each coil was paired with an individual capacitor. The capacitance value was chosen from a large set of available values so that the resulting resonance frequency of the parallel circuit is within the measurement range and the highest sensitivity, determined experimentally, is achieved.

As before, the transmission channel attached to the coils, this time operated in a parallel circuit, is first filled with water and

Table 2: Evaluation results for the coils both regarding inductance shift, measured with an impedance analyser at 500 kHz, and sensitivity in the overall setup. Each coil was paired with an optimal capacitor (determined experimentally) for operation with the inductive sensor. The highest sensitivity of a planar coil is achieved for coil 'B'.

Coil	Isolated inductance shift			Performance in complete setup			
	L (μH)	ΔL (nH)	$\Delta L/L$	Capacitor (pF)	f_{res} (MHz)	Δf_{res} (Hz)	Sensitivity (dB)
Ref	2.20	1.06	0.048 %	4700	3.21	-3176	125.0 $\pm$ 2.2
A	313.0	135	0.043 %	68	0.847	-189	77.0 $\pm$ 10.9
B	1345	1150	0.086 %	4.7	0.592	-256	99.0 $\pm$ 7.9
C	77.9	50	0.064 %	22	2.318	-402	77.0 $\pm$ 17.9
D	239.6	175	0.073 %	68	0.966	-202	91.0 $\pm$ 12.1
E	55.2	41.1	0.075 %	47	2.377	-495	80.0 $\pm$ 6.2
F	8.7	4	0.046 %	220	3.556	-400	89.0 $\pm$ 10.4

then flushed with a solution of MNPs. The change in resonance frequency is then detected using the LDC1614 inductive sensor. The sensitivity is then calculated as a signal-to-noise ratio (SNR) by determining the signal amplitude produced by the MNPs and the baseline noise value. The measurement was performed ten times and the average results were calculated.

The used optimal capacitor values, the determined resonance frequency shift Δf_{res} and the calculated sensitivity (SNR) is shown in Table 2. Although a good sensitivity is achieved for all coils, the planar coils do not reach the value of the reference coil. The highest sensitivity is achieved with the planar coil 'B'.

4.3 Placement of the Transmission Channel

Different positions of the transmission channel, relative to the planar coil were also evaluated for the coil 'B', that performed best regarding sensitivity. To this end, the transmission channel was placed with an offset of 0 mm to 20 mm from the center of the coil to then repeat the SNR measurement.

Fig. 3 shows the average sensitivity measured, dependant on the offset from the center of the coil, and the corresponding standard deviation. As the transmission channel moves away from the center of the coil, the sensitivity initially increases and then continually decreases and finally increases shortly before the edge of the coil. The short increase and the final increase is due to the maximal magnetic field density observed at the inner and outer edges of the coil in the simulation (see Fig. 2). The decrease corresponds to a reduced field density towards the outside of the planar coil.

The measured sensitivity values show a high deviation. This is due to a strong dependency of the observed noise power on the environment conditions. More accurate measurements may be achieved in future by shielding the sensor coil using material with a high permeability.

4.4 Impulse Response

The sensor response to a burst of MNPs, injected into the transmission channel, is especially relevant for a data transmission scenario. We therefore recorded the response for ten individual pulses using both the reference coil and the planar coils 'B' and 'D' to determine an average response shape for comparison. Each response was measured for an injected volume of 20 μL at a distance of 5 cm from the

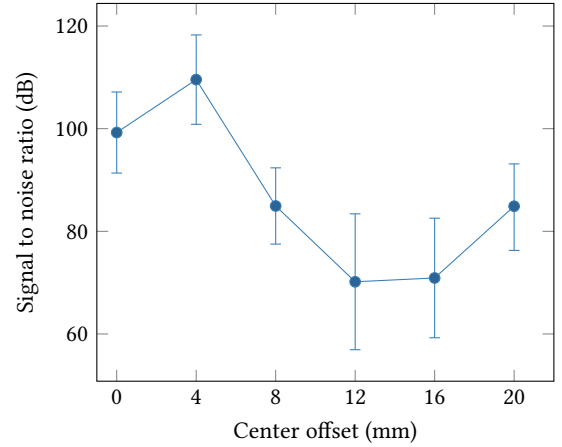


Figure 3: Evaluation regarding the impact of sensor placement on sensitivity. The SNR was measured for six different positions with the planar coil that performed best in the previous results. The highest sensitivity is achieved close to the inner edge of the coil with a decay of SNR as the distance increases.

point of injection. To calculate the average response, the individual pulses are synchronised by maximising the cross-correlation function.

Fig. 4 shows the recorded impulse responses. The normalised response for the reference coil shows less noise than that of the planar coils. This corresponds to the higher sensitivity. As the coil 'B' has more than double the size of the reference coil, the impulse response observed for that planar coil is significantly broader. Coil 'D', which is the planar coil with the next best sensitivity but is significantly smaller than coil 'B', has an impulse response similar to the reference coil.

The falling edge for coil 'B' is larger than the rising edge, causing a negative resonance shift after the impulse with a slow return to zero. A possible explanation is that a negative resonance shift, caused by eddy currents in the magnetic nanoparticle solution, becomes dominant for the large planar coil at low particle concentrations. However, further investigation in both simulation and

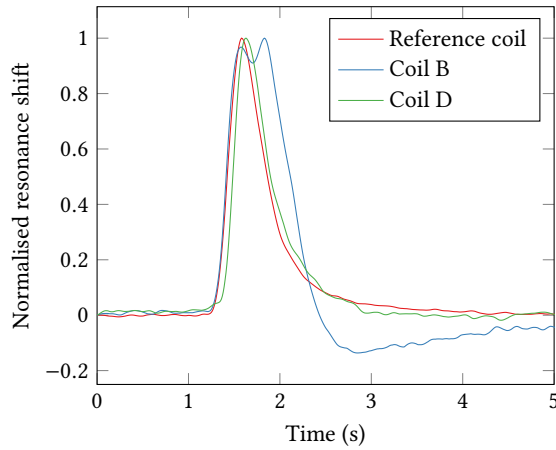


Figure 4: Average impulse response measured in the testbed setup using the planar coils 'B' and 'D', and the reference coil for a transmission distance of 5 cm and an injection volume of 20 μ L. The impulse response for coil 'D' is similar to that of the reference coil. Due to the coil dimensions, the response for coil 'B' is broader.

experimentation is necessary for a conclusive explanation, which we will perform in future work.

4.5 Data Transmission

In the molecular communication context, successful use of the receiver setup for data transmission is essential. Similar to previous publications [4, 5], we will therefore inspect a sample transmission with on-off-keying using a sequence of 18 binary symbols

$$100101\ 100000\ 110100, \quad (2)$$

corresponding to the text sequence 'FAU' according to the compact alphabet encoding presented in [4]. Each '1' was encoded by injecting 20 μ L of the MNP suspension into the transmission channel, no injection was performed for a '0'. The symbol interval was chosen as 1 s.

Fig. 5 shows the recorded resonance shift, measured at the receiver using the planar coil 'D'. Each transmitted '1' is clearly visible as a peak in the resonance shift. A simple threshold is sufficient to correctly decode the transmitted signal. A further increase of the data rate using shorter symbol intervals and more complex symbols (such as with concentration shift keying) is possible (see [2, 5] for some approaches).

5 CONCLUSION AND OUTLOOK

Six different planar coils were evaluated regarding their performance for the detection of MNPs in the context of a molecular communication testbed. In regard to sensitivity, none of the planar coils exceeded the value achieved for a wire-wound coil used as reference. However, good results, also demonstrated as average impulse responses for selected coils, could be achieved. The inductance shift of the coils was also investigated independently from the sensor device, showing consistent results in the range of 0.04 % to 0.09 %. The complete setup was tested using a simple data sequence.

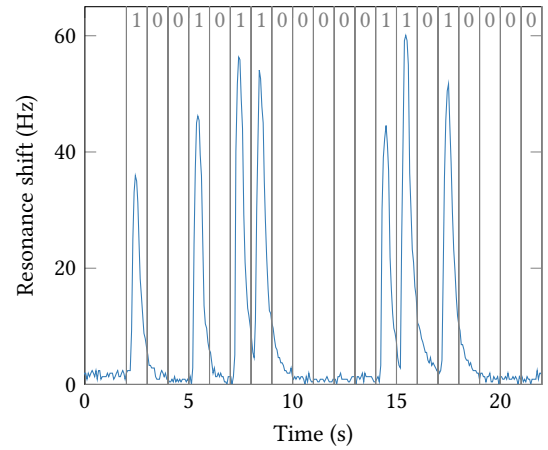


Figure 5: Sample data transmission with 18 binary symbols measured with the planar coil 'D'. One symbol was transmitted per second, denoted by the grey lines. Each interval is annotated with the transmitted symbol. The transmitted sequence can be decoded correctly with a simple signal threshold.

For the first time, planar coils were successfully applied to detect MNPs in a communication setup. In comparison to previous approaches with wire-wound coils, a planar coil sensor is easier to apply to a real-life transmission channel. Furthermore, the used planar coils are commercially available and therefore allow for a cost-efficient sensor implementation.

In future, planar coils may be used to achieve more compact sensor designs. Additionally, using an array of multiple coils could enable a more localised detection of MNPs, such as sensing of a radial MNP distribution in the transmission channel. Such a two-dimensional sensor could be used to further investigate flow behavior in testbeds and application scenarios, or to increase the transmission data rate by introducing spatial modulation.

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